

The MediaTek logo consists of the word "MEDIATEK" in white, uppercase, sans-serif font, centered within an orange parallelogram.

MT8676 Hypervisor SDCard User Manual

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Version History

Version	Date	Author	Description
1.0	2024-10-28	Andy-Id Lu	Official release
1.1	2024-11-18	Andy-Id Lu	Added support for Yocto-tbox UOS

Table of Contents

Version History	2
Table of Contents	3
List of Figures	3
List of Tables	3
1 SDCard	4
1.1 Overview	4
1.1.1 Brief Introduction	4
1.1.2 Abbreviations	4
1.2 Architecture/Process Overview	4
1.2.1 SDCard Introduction	4
1.2.2 MT8676 SDCard Feature	5
1.3 Configuration/Customization Guideline	5
1.3.1 Yocto SOS Passthrough	5
1.3.2 UOS Passthrough	8
1.4 Frequently Asked Questions/Troubleshooting	10
1.4.1 SDCard Not Recognized, VDD Voltage Could Not Be Measured	10
1.4.2 SDCard Is Recognized While Power On with Card but Failed While Hot-Plug	10
Exhibit 1 Terms and Conditions	11

List of Figures

Figure 1-1. Initialization flow of the UHS-I card	5
Figure 1-2. DTS node for SDCard	6
Figure 1-3. Pinctrl node of SDR104 mode	7
Figure 1-4. Power configuration of SDCard in the dts of Yocto SOS	8
Figure 1-5. Power configuration of SDCard in the dts of UOS	9
Figure 1-6. Remove the corresponding interrupt number on SOS	9
Figure 1-7. Add the corresponding interrupt number on UOS	9
Figure 1-8. DWS setting for detect pin	10

List of Tables

Table 1-1. Abbreviations	4
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1 SDCard

1.1 Overview

1.1.1 Brief Introduction

This Chapter introduces the hardware features, software configuration, and function of the SDCard controller. It also includes the debugging methods for common problems.

1.1.2 Abbreviations

Table 1-1. Abbreviations

Abbreviations	Explanation
DDR50	Double Data Rate up to 50MB/s@50MHz
MMC	MultiMedia Card
SDR104	Signal Data Rate up to 104MB/s@208MHz
SDR12	Signal Data Rate up to 12.5MB/s@25MHz
SDR25	Signal Data Rate up to 25MB/s@50MHz
SDR50	Signal Data Rate up to 50MB/s@100MHz
UHS-I	Ultra High Speed Phase I card

1.2 Architecture/Process Overview

1.2.1 SDCard Introduction

SDCard is a new generation of high-speed storage device based on semiconductor flash memory and developed from MultiMedia Card (MMC) format. It has high memory capacity, fast data transfer rate, great mobility flexibility, and good security, which is widely used on portable devices. In the SD3.0 protocol, theoretical maximum capacity of SDCard could be up to 2TB, theoretical maximum speed of read/write could be up to 104MB/s.

The major pins and their functions are described as below:

- (1) CLK: The clock signal, host controller or SDCard transmits one cmd/data bit at each clock cycle, up to 208MHz in UHS-I speed mode;
- (2) CMD: Command and response shared pin, command is transmitted from the host controller to the SDCard, and response is from the SDCard to the host controller;
- (3) DAT0~3: Data lines, data could be transmitted from SDCard to host controller (read), and could also be transmitted from host controller to SDCard (write);
- (4) VDD: Power supply pin of SDCard, which is normally configured to 3.3V voltage, the scope specified in the SD protocol is 2.7V~3.6V;
- (5) CD: The detection pin of the SDCard normally achieves GPIO level variation when the card is inserted or removed through the mechanical structure of SDCard holder.

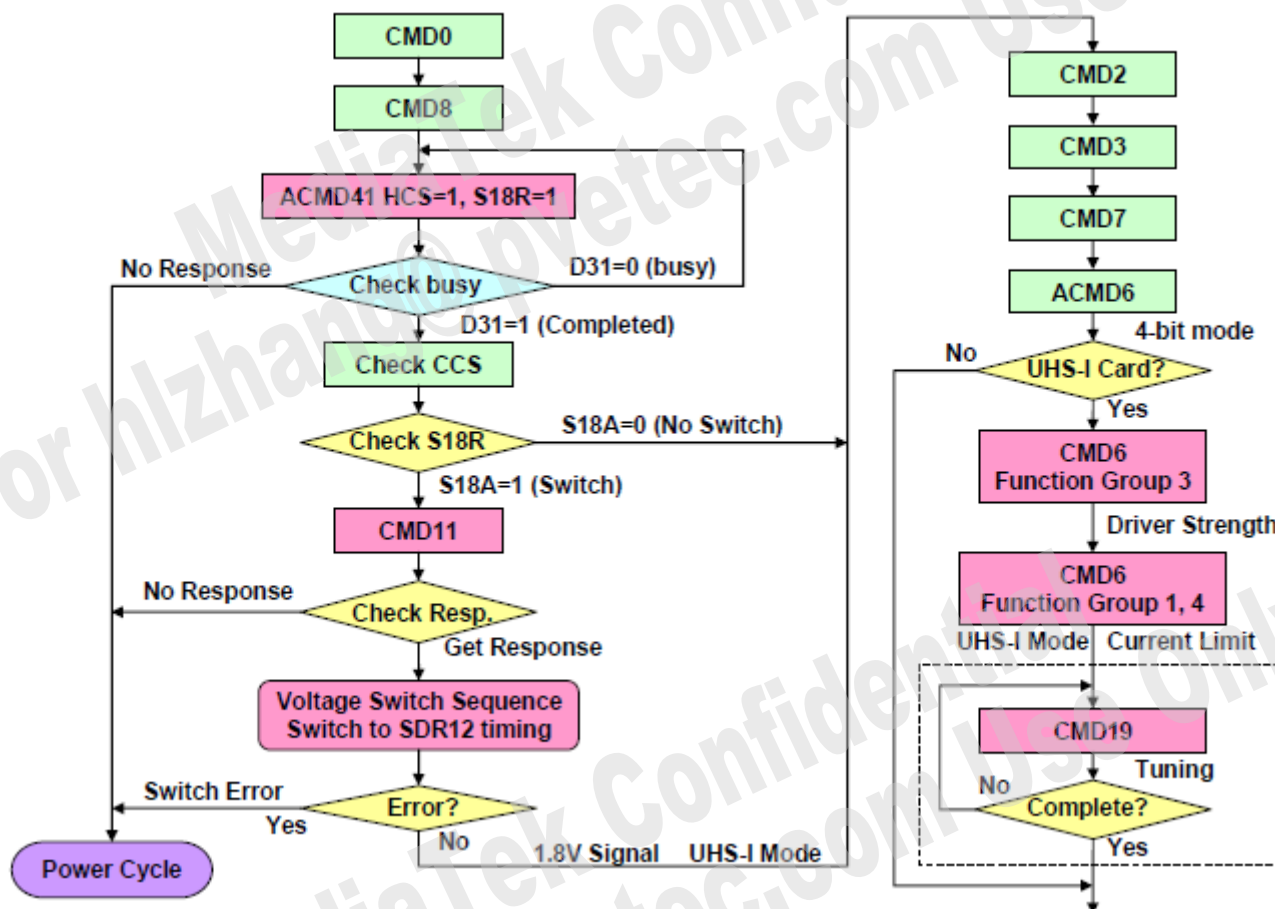


Figure 1-1. Initialization flow of the UHS-I card

1.2.2 MT8676 SDCard Feature

- (1) Compatible with SD3.0 protocol standards
- (2) Support Basic DMA and Descriptor DMA modes
- (3) Support Bus speed mode: Default Speed/High Speed/SDR12/SDR25/SDR50/SDR104/DDR50
- (4) Support 1/4-bit bus width

1.3 Configuration/Customization Guideline

1.3.1 Yocto SOS Passthrough

1.3.1.1 Kernel Config

In the configuration file:

meta/meta-mediatek-mt8676-hyp/recipes-kernel/linux/files/auto8676p1_64_hyp_defconfig

- (1) Enable SDCard support

CONFIG_MMC = y

- (2) Enable MediaTek host driver support

CONFIG_MMC_MTK_PRO = m

1.3.1.2 DTS Node

```

10 &mmc1 {
11     host-index = <1>;
12     status = "okay";
13     pinctrl-names = "default", "state_uhs";
14     pinctrl-0 = <&mmc1_pins_default>;
15     pinctrl-1 = <&mmc1_pins_uhs>;
16     bus-width = <4>;
17     max-frequency = <200000000>;
18     ocr-voltage = <0x00070000>;
19     cap-sd-highspeed;
20     sd-uhs-sdr12;
21     sd-uhs-sdr25;
22     sd-uhs-sdr50;
23     sd-uhs-sdr104;
24     sd-uhs-ddr50;
25     cd-debounce-delay-ms = <0>;
26     cd-gpios = <&pio_33 GPIO_ACTIVE_LOW>;
27     vmmc-supply = <&mt6373_vmch_eint_high>;
28     vqmmc-supply = <&mt6373_vmc>;
29     no-mmc;
30     no-sdio;
31 };

```

Figure 1-2. DTS node for SDCard

- (1) SD2.0 card needs to configure "cap-sd-highspeed", SD3.0 ultra-high speed card needs to configure "sd-uhs-xxx";
- (2) SD driving strength could be configured in the pinctrl node of the corresponding mode, such as in the SDR104 mode shown as [Figure 1-3](#);

```

74      mmc1_pins_uhs: mmc1@0{
75          pins-cmd-dat {
76              pinmux = <PINMUX_GPIO74_FUNC_MSDC1_DAT0>,
77                      <PINMUX_GPIO75_FUNC_MSDC1_DAT1>,
78                      <PINMUX_GPIO76_FUNC_MSDC1_DAT2>,
79                      <PINMUX_GPIO77_FUNC_MSDC1_DAT3>,
80                      <PINMUX_GPIO73_FUNC_MSDC1_CMD>;
81          input-enable;
82          drive-strength = <3>;
83          bias-pull-up = <MTK_PUPD_SET_R1R0_01>;
84      };
85
86      pins-clk {
87          pinmux = <PINMUX_GPIO72_FUNC_MSDC1_CLK>;
88          drive-strength = <3>;
89          bias-pull-down = <MTK_PUPD_SET_R1R0_10>;
90      };
91  };

```

Figure 1-3. Pinctrl node of SDR104 mode

- (3) SDCard detect pin could be configured by “cd-gpios”, GPIO_ACTIVE_LOW represents low card insertion level, and GPIO_ACTIVE_HIGH represents high card insertion level;
- (4) The configurations of “vmmc-supply” and “vqmmc-supply” are according to actual used SDCard VDD power and Host I/O power. If there is a need to use fast-power-off (VMCH hardware power off when SDCard is pulled out) function, “vmmc-supply” should be configured as &mt6373_vmch_eint_high (match with GPIO_ACTIVE_LOW of “cd-gpios”) or &mt6373_vmch_low (match with GPIO_ACTIVE_HIGH of “cd-gpios”); and if there is no need to use fast-power-off function, “vmmc-supply” should be configured as &mt6373_vmch.

1.3.1.3 KO Table

Add host driver ko to the following ko table path, “ramdisk” is configured in the column three, which means that ko would be installed into initramfs.

meta/meta-mediatek-mt8676-hyp/recipes-kernel/linux/ko_order_table/

auto8676p1_64_hyp/ko_order_table.csv:

```
139 mtk-mmc.ko,../kernel_device_modules-6.1/drivers/mmc/host/mtk-mmc.ko,ramdisk,Y,Y,user/userdebug/eng
```

1.3.1.4 Device Mounting

The SDCard is mounted to the share folder on the Yocto SOS side, and made multi-system accessible through virtio-fs. For example, to implement dual-system access on the Android-IVI UOS side, the reference for the mount command is as follows:

```
mkdir -p /data/share/media/sdcard
mount -t ext4 /dev/mmcblk0p1 /data/share/media/sdcard
```

The corresponding access path on the Android UOS side is /data/vendor/share/media/sdcard.

1.3.2 UOS Passthrough

1.3.2.1 Kernel Config

The configuration method remains identical to that in [Section 1.3.1.1](#), and it is already configured by default on Android-IVI.

File location in Yocto-tbox: meta/meta-mediatek-mt8676-hyp/recipes-kernel/linux/linux-mtk-extension-uos/auto8676p1_64_uos_tbox_defconfig

1.3.2.2 DTS Node

The configuration method remains identical to that in [Section 1.3.1.2](#).

1.3.2.3 KO Table

Add host driver ko to the ko table file.

File location in Android-IVI: device/mediateksample/auto8676p1_64_bsp_vm/ko_order_table.csv

File location in Yocto-tbox: meta/meta-mediatek-mt8676-hyp/recipes-kernel/linux/ko_order_table/auto8676p1_64_uos_tbox/ko_order_table.csv

```
123 mtk-mmc.ko,../../kernel_device_modules-6.1/drivers/mmc/host/mtk-mmc.ko,ramdisk,Y,Y,user/userdebug/eng
```

1.3.2.4 Configuration of Other Related Resources

(1) Power virtualization support

- Add support for the two power supplies of the SDCard in the “hyp_regulator_host” of the dts file meta/meta-mediatek-mt8676-hyp/recipes-kernel/linux/files/auto8676p1_64_hyp.dts on the Yocto SOS side.

```
1064 1064 &hyp_regulator_host {
1065 1065     status = "okay";
1066 -     regulator_count = <4>;
1066+     regulator_count = <6>;
1067 1067
1068 1068     reg1-supply = <&mt6373_vbuck2>;
1069 1069     reg2-supply = <&mt6363_vsnam_mdfe>;
1070 1070     reg3-supply = <&mt6373_vtp>;
1071 1071     reg4-supply = <&mt6319_6_vbuck1>;
1072+     reg4-supply = <&mt6373_vmc>;
1073+     reg5-supply = <&mt6373_vmch_eint_high>;
1072 1074 };
```

Figure 1-4. Power configuration of SDCard in the dts of Yocto SOS

- Remove the settings for the two power supplies of the SDCard in the dts on the UOS side.

```

1828 -
1829 - &mt6373_vmc {
1830 -     regulator-min-microvolt = <0>;
1831 - };
1832 1828
1833 1829 &mt6373_vaud18 {
1834 1830     regulator-min-microvolt = <0>;
1835     * 18 hidden lines
1853 1849 &mt6373_vefuse {
1854 1850     regulator-min-microvolt = <0>;
1855 1851 };
1856 -
1857 - &mt6373_vmch_eint_high {
1858 -     regulator-min-microvolt = <0>;
1859 - };

```

Figure 1-5. Power configuration of SDCard in the dts of UOS

(2) Interrupt virtualization support

The configuration in thyp-sdk should be modified to include support for interrupt number 166 (which requires an increase of 32 to reach 198) for the SDCard host on the UOS side.

- Remove the interrupt number in sos_mt8676.json

```

11 11 "vgic": {
12 12     "gicd_paddr": "0xc400000",
13 13     "gicr_paddr": "0xc440000",
14 14     "percpu_interrupt": [
15 15         "23"
16 16     ],
17 17     "interrupt": [
18 18         "32-644",
19+ 18+     "32-197",
20+ 19+     "199-644",
19 20     "647-920",
20 21     "922",

```

Figure 1-6. Remove the corresponding interrupt number on SOS

- Add the interrupt number in uos_alps_pv8676.lua(Android-IVI) or uos_tbox_pv8676.lua(Yocto-tbox)

```

109 109 uos_config:setVgicIts(1, 0x91000000)
110 110 -- interrupt
111 - interrupt = {70, 634, 635, 645, 646, 921, 923, 924, 929}
111+ interrupt = {70, 198, 634, 635, 645, 646, 921, 923, 924, 929}
112 112 for i = 1, #interrupt do
113 113     uos_config:setVgicIrq(interrupt[i])
114 114 end

```

Figure 1-7. Add the corresponding interrupt number on UOS

1.4 Frequently Asked Questions/Troubleshooting

1.4.1 SDCard Not Recognized, VDD Voltage Could Not Be Measured

- (1) Check the correction of Kernel config and DTS according to [Section 1.3](#);
- (2) If VDD supplied power is MT6373, and detect pin is connected to SD_DET pin of MT6373, please check if the configured power node of “vmmc-supply” is matched with the polarity of detect pin;
- (3) If the result of step (2) is okay, please configure “vmmc-supply” as &mt6373_vmch and check VDD voltage, correct voltage represents there is a problem in fast-power-off function, submit PMIC issue to MediaTek;
- (4) If VDD could not power up yet in step (3), please catch Kernel log and submit SDCard issue to MediaTek.

1.4.2 SDCard Is Recognized While Power On with Card but Failed While Hot-Plug

- (1) Check the correction of “cd-gpios” configuration in DTS according to the [Section 1.3.1.2](#);
- (2) If DTS configuration is okay, check the correction of the GPIO configuration for detect pin in the file `src/devtools/dct/dws/mt6897/${PROJECT}.dws` (Yocto path) or `vendor/mediatek/proprietary/tools/dct/dws/mt6897/${PROJECT}.dws` (Android path), refer to the configuration shown in [Figure 1-8](#):

```

430 <gpio33>
431 <eint_mode>true</eint_mode>
432 <def_mode>0</def_mode>
433 <inpull_en>true</inpull_en>
434 <inpull_selhigh>true</inpull_selhigh>
435 <def_dir>IN</def_dir>
436 <out_high>>false</out_high>
437 <varName0>GPIO_SDHC_EINT_PIN</varName0>
438 <smt>true</smt>
439 <ies>true</ies>
440 </gpio33>

```

Figure 1-8. DWS setting for detect pin

- (3) If the DTS and DWS configuration are all okay but SDCard could not be recognized yet while hot-plug, please measure the level of the detect pin in the state of card inserting and removing and submit the SDCard issue to MediaTek if the result of measurement meets expectations.

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